AI 321 MACHINE LEARNING FOR SEQUENTIAL DATA

A. Course Catalogue Description

1. **Course Number:** AI 321
2. **Course Title:** Machine Learning for Sequential Data
3. **Course Description:** Design and development of applications of various machine learning algorithms on predictive maintenance and energy sequential data
4. **Prerequisite:** AI 221 Classical Machine Learning/equivalent
5. **Semester Offered:** First / Second Semester
6. **Course Credit:** 3 units
7. **Number of Hours:** 3 hours
8. **Meeting Type:** Lecture
9. **Course Goals:** The course intends to teach the students on how to process sequential data using artificial intelligence / machine learning towards achieving engineering research goals.

B. Rationale

Sequential data sets are found ubiquitously in many areas of engineering, as they often arise through measurements collected as a time series. A few examples include data sets for electrical load forecasting, renewable energy resource prediction, predictive maintenance, battery life modelling, equipment degradation modelling, speech recognition, and video processing. In chemometrics, data such as from GC-MS (Gas chromatography - mass spectrometry), FTIR (Fourier-transform infrared spectroscopy) absorbance data, and NMR data (Nuclear magnetic resonance) can also be treated as a time series. There are specialized techniques for performing machine learning from these sequential data, which the students must learn and apply in real-world scenarios.

C. Course Content

Lecture Topics	No. of Hours
I. Introduction to Time Series Analysis A. Definition B. Autocorrelation C. Seasonality D. Trends	3
II. Univariate Time Series Modelling A. AR, ARMA, ARMAX B. ARIMA, ARIMAX C. Other Polynomial Models (Box-Jenkins, Transfer Functions) D. Gaussian Process AR / ARX Models	6
III. Multivariate Time Series Modelling A. VAR, VARMA B. Co-integration	3
IV. Smoothing and Filtering A. Moving average B. Exponential smoothing, EWMA C. Finite Impulse Response (FIR) D. Kernel smoothing	3
V. Feature Extraction from High-frequency Signals A. Power Spectral Density	6

B. Wavelet Transform (Discrete, Continuous) C. Empirical Mode Decomposition D. Hilbert-Huang Transform	
VI. Clustering Time-Series Data A. Notions of Similarity between Time Series B. Dynamic Time Warping C. Edit Distances D. Clustering using ML methods	3
VII. State-space Models and Hidden Markov Models A. Linear Dynamics B. State-space Models C. Hidden Markov Models D. Viterbi Algorithm	3
VIII. System Identification: Prediction Error Methods A. Recurrent NNs B. LSTMs C. Neuro-Fuzzy Models D. Deep Learning models	6
IX. System Identification: Subspace Identification A. Hankel Matrices B. N4SID C. CVA	3
X. Case studies for System Identification A. Evaporator System B. Narendra-Li C. CSTR	3
XI. Forecasting for Predictive Maintenance A. Faults and Failures B. Health Indicators C. Remaining Useful Life D. Degradation Models	3
XII. Forecasting for Energy and Environmental Applications A. Solar, Wind, Load Demand Forecasting B. Spatio-temporal Data	3
XIII. Other Issues in Time Series Data A. Journal Readings B. Latest Research	3
TOTAL NUMBER OF HOURS	**48**

D. Course Coverage

Week	Learning Outcomes	TOPIC	ESSENTIAL/ KEY QUESTIONS	Suggested Teaching and Learning Activities	Suggested Assessment Tools	Course Readings / Learning Resources
1	• Determine the nature of time-series data • Compute correlations, trends, and seasonality indices from time-series data	**I. Introduction to Time Series Analysis** A. Definition B. Autocorrelation C. Seasonality D. Trends	What are examples of time series data in engineering? How are they generated? How to compute measures such as autocorrelation, seasonality, trends?	Lecture	Quiz	Box et al. (2015) Hyndman and Athanasopoulos (2018)
2-3	• Fit univariate time-series data to simple polynomial autoregressive models • Implement Bayesian time-series modelling using Gaussian processes	**II. Univariate Time Series Modelling** A. AR, ARMA, ARMAX B. ARIMA, ARIMAX C. Other Polynomial Models (Box-Jenkins, Transfer Functions) D. Gaussian Process AR / ARX Models	What are the different time series models? What are the models under the generalized polynomial class (AR, ARMA, ARMAX, ARIMA, BJ, Transfer functions)? How to fit a Gaussian process AR model?	Lecture, Software Demo	Machine Exercises	Box et al. (2015) Hyndman and Athanasopoulos (2018)
4	• Fit multivariate time-series data to vector autoregressive models • Understand concepts of co-integration in multivariate time-series.	**III. Multivariate Time Series Modelling** A. VAR, VARMA B. Co-integration	How to fit multivariate time series models such as VAR, VARMA, VARMAX, and state-space models? What is co-integration?	Lecture, Software Demo	Machine Exercises	Box et al. (2015) Hyndman and Athanasopoulos (2018)
5	• Perform smoothing and filtering techniques • Apply appropriate pre-processing techniques on time-series data	**IV. Smoothing and Filtering** A. Moving average B. Exponential smoothing, EWMA C. Finite Impulse Response (FIR) D. Kernel smoothing	What are smoothing and filtering? How to apply moving average and exponential smoothing? What are kernel smoothing methods?	Lecture, Software Demo	Machine Exercises	Verhaegen and Verdult (2007)
6-7	• Apply feature extraction techniques on high-frequency signals.	**V. Feature Extraction from High-frequency Signals** A. Power Spectral Density	How to extract features from time-series data? How to extract Power Spectral Density features? Wavelet transform features. What can we accomplish from	Lecture, Software Demo	Machine Exercises	

	● Utilize features rather than the original time-series towards machine learning.	B. Wavelet Transform (Discrete, Continuous) C. Empirical Mode Decomposition D. Hilbert-Huang Transform	using machine learning on those features rather than the original time-series directly?			
8	● Use time-series features for clustering the data ● Analyze similarity between time-series using various metrics such as edit distance.	**VI. Clustering Time-Series Data** A. Notions of Similarity between Time Series B. Dynamic Time Warping C. Edit Distances D. Clustering using ML methods	How to determine similarity between time-series data? How to use different similarity measures for clustering / feature extraction, etc.?	Lecture, Software Demo	Machine Exercises	
9	● Simulate the properties of models with internal dynamics. ● Apply hidden markov models and state-space models in describing internal dynamics.	**VII. State-space Models and Hidden Markov Models** A. Linear Dynamics B. State-space Models C. Hidden Markov Models D. Viterbi Algorithm	What are the different ways to model linear dynamical systems? What are state-space models? What are hidden markov models? What are their properties?	Lecture, Software Demo	Machine Exercises	
10-11	● Simulate using neural nets, fuzzy models, and deep learning models. ● Understand the limitations of deep learning for time-series modelling.	**VIII. System Identification: Prediction Error Methods** A. Recurrent NNs B. LSTMs C. Neuro-Fuzzy Models D. Deep Learning models	What is system identification? How to train recurrent neural networks for time series? How to train gated recurrent units and long short-term memory networks? How to leverage appropriate deep learning methods for modelling?	Lecture, Software Demo	Machine Exercises	
12	● Simulate data using subspace identification methods. ● Improve system identification using regularization and kernel designs.	**IX. System Identification: Subspace Identification** A. Hankel Matrices B. N4SID C. CVA	What are subspace identification methods (SIMs)? What are the N4SID methods? How to apply regularization and kernel methods in SIMs?	Lecture, Software Demo	Machine Exercises	
13	● Apply system identification for modelling various real-world processes.	**X. Case studies for System Identification** A. Evaporator System B. Narendra-Li C. CSTR	What are well-known case studies in system identification? How to use state-space models in simulation and prediction?	Case Studies	Seatwork, Quiz ML Project Proposal	Research Papers

14	● Identify basic activities in predictive maintenance, especially forecasting ● Model the degradation of usual systems (batteries, reactors, etc.)	**XI. Forecasting for Predictive Maintenance** A. Faults and Failures B. Health Indicators C. Remaining Useful Life D. Degradation Models	What is predictive maintenance? How to do forecasting of equipment health indicators and degradation? How to fuse different data sources into one model?	Lecture, Case Studies	Seatwork, Quiz	
15	● Forecast data sets in renewable energy ● Use spatio-temporal analytics to describe and model physical phenomena.	**XII. Forecasting for Energy and Environmental Applications** A. Solar, Wind, Load Demand Forecasting B. Spatio-temporal Data	What are the typical time series in renewable energy? What can we learn from solar, wind, biomass, etc. spatio-temporal data? How to model environmental variables?	Case Studies	Machine Exercises	Research Papers
16	● Identify other issues in time series data	**XIII. Other Issues in Time Series Data** A. Journal Readings B. Latest Research	How to handle other issues in time series such as missing data, uncertainty, etc.? What are the latest published papers in the field of time series analysis?	Journal Presentation	Journal Critique ML Project	Research Papers

E. Course Requirements

Two long Exams:	Midterm and Finals (40%)
Machine Problems	3-5 Machine Problems (35%)
ML Project	Course Project (20%)
Journal / Conference Paper Reading	Report / Critical Analysis (5%)

F. REFERENCES:

Statistical and Adaptive Signal Processing: Spectral estimation, signal modeling, adaptive filtering and array processing, D. G. Manolakis, Vinay G. Ingle, S. M. Kogon, 3rd ed, 2005 Artech House Inc.

Box et al. (2015). Time Series Analysis: Forecasting and Control, 5th Ed. Wiley.

Holton, T. (2021). Digital Signal Processing: Principles and Applications. 1st Ed. Cambridge University Press.

Digital Signal Processing: Principles, Algorithms and Applications, John G. Proakis, Dimitri G. Manolakis, 3rd ed, Prentice Hall Inc. (1996)

Hyndman and Athanasopoulos (2018). Forecasting: Principles and Practice. 2nd Ed. OTexts: Melbourne, Australia : https://otexts.com/fpp3/

Linear System: A State Variable Approach, Raymond A. DeCarlo, 1989, Prentice Hall

Probability, Random Variables and Random Signal Principles, Peyton Z. Peebles and Bertram Shi, 2015 McGraw Hill.

G. Supplementary References

Atienza R (2020). Advanced Deep Learning with Tensorflow 2 and Keras. Packt Pub.

Bishop (2006). Pattern Recognition and Machine Learning. Springer.

James et al. (2021). Introduction to Statistical Learning with Applications in R. 2nd Ed.

Jalao, E. (2021) Data Analytics For Engineers and Scientists, UP Press

Ljung (1987). System Identification: Theory for the User. Prentice Hall.

Murphy, K. (2021). Probabilistic Machine Learning: An Introduction. MIT Press.

Nate Silver (2015). The Signal and the Noise: Why So Many Predictions Fail--but Some Don't, Penguin Books.

Nelles. (2001) Nonlinear System Identification. Springer-Verlag Berlin.

Rasmussen and Williams (2006), Gaussian Processes for Machine Learning, MIT Press.

Russell & Norvig (2020). Artificial Intelligence: A Modern Approach. 4th Ed. Prentice Hall.

Verhaegen and Verdult. (2007) Filtering and System Identification: A Least Squares Approach. Cambridge University Press.